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Long-duration flight leads to sensorimotor problems which can be critical during landing, rendezvous, and operations on other planetary surfaces. While specific sensorimotor effects have been identified, it is not known which ones have the most adverse impact, or how best to assess them and apply appropriate rehabilitation procedures. NASA’s current goal in addressing this situation is to develop a means to assess sensorimotor function rapidly with a portable device, so that an astronaut can make a determination as to whether or not he or she is impaired enough to affect mission safety or success before undertaking a demanding task (piloting, landing, reentry, egress, EVA, tele-operation, etc.). Accordingly, the goal of this project is to develop a portable hand-held device that will allow a single crewmember to assess his/her sensorimotor function in no more than 20 minutes. This is to be accomplished with a judicious choice of which sensorimotor functions to assess, and careful design to obtain the maximum data in the minimum time. The device and procedures being developed to meet these requirements are based on a notebook computer, attached to a joystick and body-mounted motion-sensor units. Through a set of simple software routines, a rapid assessment of

Task Description:	sensorimotor capabilities will be made. The final device will be small, require little power and space, and provide what is essentially a self-contained sensorimotor lab/clinic. We term this device and its embedded software the Sensorimotor Assessment and Rehabilitation Apparatus (SARA). SARA will be designed to measure, initially, the following functional behaviors that are relevant to safe and effective functioning and are also affected by long-duration space flight: 1. Vestibulo-ocular function during pitch head movements (eye movements in response to head movements, a direct measure of vestibular function and canal-otolith integration, and highly relevant for gaze stabilization). Originally we had intended to use dynamic visual acuity (DVA) for this test, but have recently developed an alternative method that uses subjective nulling of apparent visual motion during head movement. We term this vestibulo-ocular nulling (VON). 2. Vestibulo-ocular function as mediated by the otolith organs. As above, we had intended to use DVA for this test, but have since been developing VON as a replacement. Given the difficulty of self-generating translational movements as stimuli for this test, we are extending VON to measure vertical skew and disconjugate torsion, which are more reliable otolith function tests. 3. Saccade targeting as a measure of spatial orientation and perception. 4. Postural control, by measuring body sway with eyes closed. This test is enhanced by having the subject make pitching head movements. 5. Locomotion, by measuring alterations in the coupling between pitching head movements and vertical body motions during walking, and by assessing variability in stride length and timing which are predictors of fall risk.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our sensorimotor assessment apparatus is based on a notebook computer, motion sensors, visual display, and joystick. This simple and portable device meets the need for rapid non-invasive screening of vestibular and sensorimotor function in settings where special expertise and equipment is not available. Applications include assessment of passengers before and after parabolic and sub-orbital flight, and assessment/rehabilitation of vestibular patients away from specialized centers.
Task Progress:	Significant effort to date has been devoted to assembling hardware and software components to implement the initial set of SARA tests. This includes an HP Slate 500 tablet PC computer, which will host all software and provide a display for saccade targeting. Also acquired is a set of Opal sensors (APDM Inc.). These are self-contained linear and angular motion sensors, which send data to a base station for storage and computer processing. Software development continues for posture and locomotion assessment with Opals, and for saccade targeting with the tablet PC. The Opal sensors have had a number of problems, including calibration and wireless communication, which have been resolved through extensive consultation with the manufacturer. Placement of sensors on the body, and the number required, will be one of the key questions addressed in our initial studies with this equipment. We finalized our protocol for parabolic-flight investigations, and obtained IRB approvals from NASA and Johns Hopkins. We also began protocol development for planned centrifuge testing at NASTAR Center, as an additional analog environment. The largest single effort has gone into the development and testing of our vestibulo-ocular nulling device (VOND). The test hardware for this device includes a biteboard, rate sensor (attached to biteboard), data acquisition card, laptop computer, and head-mounted display. The computer generates a visual target whose position on the display is determined by the movement of the head and a motion-gain value set by the subject. The subject controls the gain of the head-motion/target-motion process through a rotary knob, adjusting gain as necessary so that the target appears stationary when the head moves (pitching head movements). If the vestibulo-ocular reflex (VOR) is miscalibrated (i.e., the eye movements do not appropriately compensate for head movements, as is the case the during g level changes), a stationary target will appear to move when the head moves. If head movement is measured in real time, and used to control target motion through a variable gain, the subject can adjust that gain so that the target appears stationary during head motion. The motion-gain setting determines to what extent the target must be moved – either more or less than the head movement – so that it matches the eye movements and appears to be stationary. If the VOR gain is too low, the eyes move less than the head (in the opposite direction), and the motion-gain setting will be less than 1.0 in order to null the perceived target movement. Likewise a VOR gain that is too high will lead to a motion-gain setting greater than 1.0. Thus the motion-gain value, set by the subject to null perceived target movement, provides a surrogate measure of VOR gain, without measuring eye movements. The effects of g level are determined by comparing responses from identical motions in 0g and 1.8g. This was tested on a set of parabolic flights in the fall of 2010. We first tested the VOND apparatus and procedures in the lab. Four normal subjects viewed the computer-generated target through convex lenses (ranging from 1.5 to 6 diopters), inducing a miscalibration in VOR gain. The motion-gain was initially set to 1.0 for each trial; subjects then adjusted the gain setting until the visual target appeared stationary while making sinusoidal head movements. Then, the initial motion-gain setting was changed by the operator to that predicted by the lens power; subjects again adjusted the gain to null perceived motion. Tests were performed in the dark. Subjects readapted back to the normal gain of 1.0 before lenses were changed for the next trial. Since convex lenses magnify the visual scene, they increase the amount by which images move across the retina as the head turns. This required subjects to have VOR gains greater than 1.0 in order to properly track a stationary target and perceive it as stationary. Thus they would have to make motion-gain settings less than 1.0 to null apparent target motion. As predicted, when lens power increased, subjects required smaller motion-gain settings in order to null perceived target motion. In addition, we found that subjects tolerated a range of motion-gain values over which the visual target appeared stationary. Flight testing was successful, although shortcomings in the current design were noted. Most notably, interaction between operator and subject was sometimes cumbersome, and this was very obvious in the parabolic-flight environment with

ambient noise and distractions from other activities. The other major finding regarding the device itself is that the task that the subject must perform is not as intuitive as we had hoped, especially in an unusual setting where g levels are changing. Subjects attempted to interpret the visual target dot on the head-mounted display as existing “beyond the display” – perhaps projected onto the opposite wall of the aircraft – but this proved difficult due to ambient lighting and glare on the display, drift of the target due to imperfect head-movement sensors, and the overriding mental set that clearly placed the target on the nearby headset. All of these issues are correctable with hardware refinements, which we intend to implement in the near future.

Even with these difficulties, we obtained quality data that demonstrate the validity of the concept. Our two experienced subjects, and one naïve subject, made consistent and reliable gain settings which we can relate to underlying sensorimotor function. Two other naïve subjects either could not perform the task due to motion sickness, or had difficulty interpreting the instructions, which provides us with important information on how to design and explain the procedures to our subjects. The main results show that gain is lower in the reduced-gravity levels due to a reduced otolith contribution, and higher in the hyper-gravity levels. In three of four subjects this is the case.

We are now modifying this device to enable similar subjective assessment of vertical skew and disconjugate torsion.

Summary

Progress, key issues, and immediate plans for each of the initial tests in our assessment battery are as follows:

1. Vestibulo-ocular nulling device (VOND). This has been tested in the lab and parabolic flight. The basic procedure has been verified, but work remains to make the test easier and more intuitive. Testing in parabolic flight indicates that g-related changes can be identified, but changes due to adaptation are smaller and therefore will provide a stronger test of the apparatus. We plan on changing the display and testing in future parabolic flights and during centrifugation.

2. Otolith testing. We now plan to modify VOND for skew and torsion assessment, which are measures of otolith function and asymmetry. This will also be validated with flight and centrifuge tests.

3. Saccade targeting. While we prefer to use saccades here to assess spatial awareness and visual reference frames, saccades present an operational problem because we wish to avoid measurement of eye movements per se and it is very difficult to stabilize a subject in parabolic flight relative to any external visual targets. To avoid both problems, we designed a simple paradigm, using the tablet PC, to assess saccade targeting in parabolic flight. This consists of showing an array of targets on the PC, which the subject holds on his or her lap, while making repeated saccades to the center target and to an external target elsewhere in the aircraft (or lab). The targets are extinguished and the saccades continue briefly, and when the saccades stop the targets reappear. By indicating which of the array of targets the subject is looking at most directly, a surrogate measure of saccade targeting is obtained. We will be testing this procedure in upcoming flights and possibly during centrifugation.

4. Posture testing will take place as planned, with the subject's feet together and eyes closed while making pitch head movements. Body sway will be recorded and compared to baseline measures. A new feature of this test is its combination with VOND testing. Since both tests make use of pitch head movements, we can save time and make the tests more functionally meaningful by measuring body sway during VOND testing.

5. Locomotion will be tested as planned, with the wireless Opal sensors while the subject makes head movements while walking. A minor question is how far a person will have to walk in order for deficits to be detected, and this will be addressed in our flight and centrifuge studies.

Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Beaton KS, Ying HS, Roberts DC, Shelhamer M. "Assessing vestibulo-ocular function without measuring eye movements." Presentation at the XXVI Bárány Society Meeting, Reykjavik, Iceland, 18-21 August 2010. XXVI Bárány Society Meeting, 2010. http://www.eventure-online.com/eventure/publicAbstractView.do?id=143979&congressId=3627 , Aug-2010
Abstracts for Journals and Proceedings	Beaton K, Shelhamer M. "Sensorimotor assessment for suborbital passengers: a recommended protocol." Presentation at the 2nd Next-Generation Suborbital Researchers Conference, Orlando FL, 27 February – 2 March 2011. 2nd Next-Generation Suborbital Researchers Conference, 2011. http://www.swri.org/9what/events/confer/nsrc/2011/index.htm ; http://www.swri.org/9what/events/confer/nsrc/2011/abstracts/life/288826892214460.pdf , Mar-2011
Abstracts for Journals and Proceedings	Shelhamer M, Beaton K. "Possible pre-flight adaptation protocols for suborbital passengers: recommendations based on pre-flight testing." Presentation at the 2nd Next-Generation Suborbital Researchers Conference, Orlando FL, 27 February – 2 March 2011. 2nd Next-Generation Suborbital Researchers Conference, 2011. http://www.swri.org/9what/events/confer/nsrc/2011/index.htm ; http://www.swri.org/9what/events/confer/nsrc/2011/abstracts/life/28851578-2214460.pdf , Mar-2011
Abstracts for Journals and Proceedings	Beaton KH, Karmali F, Shelhamer M. "Gravity dependence of the pitch vestibulo-ocular reflex." Presentation at the 8th Symposium on the Role of the Vestibular Organs in Space Exploration, Houston TX, 8-10 April 2011. 8th Symposium on the Role of the Vestibular Organs in Space Exploration, 2011. http://www.dsls.usra.edu/meetings/IAA/vestibular/index.shtml ; http://www.dsls.usra.edu/meetings/IAA/vestibular/pdf/3034.pdf , Apr-2011
Abstracts for Journals and Proceedings	Shelhamer M, Beaton KH. "Pre-flight adaptation protocols for suborbital flight. " Presentation at the 8th Symposium on the Role of the Vestibular Organs in Space Exploration, Houston TX, 8-10 April 2011. 8th Symposium on the Role of the Vestibular Organs in Space Exploration, 2011. http://www.dsls.usra.edu/meetings/IAA/vestibular/index.shtml ; http://www.dsls.usra.edu/meetings/IAA/vestibular/pdf/3061.pdf , Apr-2011

Abstracts for Journals and Proceedings	Beaton KH, Shelhamer M. "Design considerations for postflight sensorimotor testing and rehabilitation." Presentation at the 18th IAA Humans in Space Symposium, Houston TX, 11-15 April 2011. 18th IAA Humans in Space Symposium, Program and Abstracts, 2011. , Apr-2011
Articles in Peer-reviewed Journals	Shelhamer M, Beaton K. "Mission to Mars: training and maintenance of sensorimotor responses considerations based on context-specific adaptation." Journal of Cosmology 2010 Oct-Nov;12:3817-24. http://journalofcosmology.com/Mars135.html , Nov-2010